

Sudden Cardiac Death and its Relation to Previously Diagnosed or Occult Cardiac Disease at Autopsy

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ABSTRACT

BACKGROUND Sudden cardiac death (SCD) prevention focuses on individuals with diagnosed disease (ie, conventional SCD risk factors [RFs]) such as reduced ejection fraction and myocardial infarction (MI). The burden of occult disease among community SCDs without diagnosed RFs is unknown and represents a target for prevention through increased detection.

OBJECTIVES This study sought to determine the sensitivity of diagnosed RFs for community sudden deaths and identify cardiac pathology, including occult MI and dilated cardiomyopathy (DCM), among community sudden deaths without diagnosed RFs.

METHODS The POST SCD (Postmortem Systematic Investigation of Sudden Cardiac Death) is a prospective countywide study using autopsy to adjudicate arrhythmic (potentially rescuable with defibrillator) or nonarrhythmic (eg, tamponade, overdose) deaths among presumed SCDs meeting World Health Organization criteria. We assessed prevalence (“sensitivity”) of diagnosed RFs (ejection fraction $\leq 35\%$, heart failure, prior MI, syncope) among arrhythmic, nonarrhythmic, and reference trauma deaths. Among arrhythmic deaths without diagnosed RFs, we assessed occult cardiac pathologies including DCM (short-axis diameter ≥ 3.5 cm and heart weight 1 SD more than expected based on sex, age, height, and weight; Z-score =1) and healed MI (histopathological evidence of healed MI).

RESULTS Of 877 presumed SCDs, 513 (58%) were autopsy-defined arrhythmic deaths, of which 166 subjects (32%) had diagnosed RFs (mean age: 64.3 years; 77% men); therefore, sensitivity of RFs for arrhythmic death was 32%. Another 159 subjects (31%) had occult MI or DCM with similar demographics (mean age: 62.6 years; 80% men) and cardiac pathologies as those with RFs, including fibrosis and coronary disease. The remaining 185 arrhythmic deaths (36%) were subjects who were younger (mean age: 56.9 years) with less cardiac pathology than arrhythmic deaths with occult MI or DCM but still had increased heart weight (Z-score: 0.9 vs 0.0), larger left ventricular diameter (2.5 cm vs 1.9 cm), and more significant coronary disease (52% vs 13%, all $P < 0.001$) but similar fibrosis (6.7% vs 6.3%) and left ventricular hypertrophy burden (57% vs 55%) than trauma deaths.

CONCLUSIONS In this 12-year countywide postmortem study, two-thirds of community arrhythmic SCDs occurred in individuals without diagnosed disease despite substantial cardiac pathology; half of these “silent” arrhythmic deaths had occult MI or DCM. Improved detection of occult cardiac disease is a critical priority to reduce community sudden deaths. (JACC. 2026;■:■-■) © 2026 Published by Elsevier on behalf of the American College of Cardiology Foundation.

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

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**ABBREVIATIONS
AND ACRONYMS****CAD** = coronary artery disease**DCM** = dilated cardiomyopathy**GDMT** = guideline-directed
medical therapy**HF** = heart failure**HFREF** = heart failure with
reduced ejection fraction**ICD** = implantable
cardioverter-defibrillator**LV** = left ventricular**LVEF** = left ventricular
ejection fraction**LVH** = left ventricular
hypertrophy**MI** = myocardial infarction**RF** = risk factor**SCD** = sudden cardiac death

The elucidation of risk factors (RFs) and conditions underlying sudden cardiac death (SCD) is crucial to identify individuals at greatest risk and deploy prevention strategies, such as implantable cardioverter-defibrillators (ICDs), *before* SCD. Decades of investigation have identified heart failure (HF) with reduced ejection fraction (HFREF), myocardial infarction (MI), and syncope as key RFs for lethal arrhythmia.¹⁻⁴ However, despite significant advances in the management of these conditions, including primary and secondary prevention of atherosclerotic cardiovascular disease and expansion of guideline-directed medical therapy (GDMT) for HF, and decreasing rates of sudden death in these populations (eg, patients with HF),⁵ SCD remains a leading cause of global mortality. Thus, multiple guidelines and expert

consensus statements have advocated for reappraisal and innovation of SCD risk stratification.^{1,4,6,7} Essential to these efforts is a better understanding of who dies suddenly and why we miss them.

Current SCD prevention strategies can only be applied to patients with diagnosed cardiovascular disease and as such address only some of the at-risk population (eg, ICDs for patients with HFREF on GDMT).^{8,9} Prior epidemiological studies have reported that upward of ~70% of all SCDs occur in individuals without clinically recognized cardiovascular disease.¹⁰⁻¹⁴ However, these data are limited by an SCD definition that presumes arrhythmic cause without autopsy¹⁵⁻¹⁹ and thus were biased by the inclusion of noncardiac sudden deaths (eg, occult overdose, stroke), which have a lower burden of cardiovascular disease.²⁰ We showed that nearly one-half of these traditionally defined SCDs (eg, World Health Organization criteria) actually have nonarrhythmic cardiac (eg, tamponade) or noncardiac causes by autopsy²⁰ thus are now considered “presumed SCDs” by new international consensus nomenclature.^{6,21} Moreover, the burden of cardiac pathology among SCDs due to “occult” pathology not recognized in routine care is unknown.

Herein, autopsy studies have the potential to shed key insights into why sudden deaths are missed. Historical exemplar autopsy studies have laid the foundation for our understanding of sudden death, demonstrating MI as the cause of ~80% of SCDs^{22,23}; however, these studies were limited by referral bias of the minority of total SCDs autopsied. A more

recent study in Northern Finland detected silent MI in 42% of autopsied SCDs due to coronary artery disease (CAD).²⁴ Finnish law enables some of the highest autopsy rates worldwide, but some SCDs “due to a known disease” may not undergo autopsy,²⁵ and therefore the occult disease rate in autopsied SCD cases may be inflated. Other contemporary autopsy studies with more comprehensive capture of cases have uncovered underlying causes of sudden deaths in young populations,²⁶ but their findings are limited to these selected population with inherently high autopsy rates.

In this update of the prospective San Francisco POST SCD (POStmortem Systematic inVeStigation of Sudden Cardiac Death) study, we sought to evaluate the burden of diagnosed and undiagnosed cardiac disease in presumed SCDs in people aged 18 to 90 years from a diverse urban U.S. population. We first identified how many victims died suddenly with diagnosed, conventional RFs including HF, MI, or syncope to assess sensitivity of current risk stratification criteria. We then evaluated the burden of occult (ie, undiagnosed) cardiovascular disease, including MI and dilated cardiomyopathy (DCM), in those without documented RFs, representing a high-risk population currently missed for SCD prevention. Finally, we compared cardiac pathologies in decedents with occult cardiac disease with those with diagnosed disease and control trauma deaths, the latter an estimate of the background population prevalence of heart disease.

METHODS

The study was approved by the Institutional Review Board of the University of California-San Francisco, with additional approval at all 10 adult hospitals and 3 emergency medical service agencies in San Francisco County to obtain medical records.

STUDY POPULATION. The complete POST SCD study protocol has been described previously.²⁰ Briefly, POST SCD is a prospective countywide study of out-of-hospital cardiac arrest²⁷ deaths in those aged 18 to 90 years in San Francisco County. The study population included all residents and inhabitants of San Francisco County, California. A single Office of the Chief Medical Examiner serves the County of San Francisco. All out-of-hospital and unattended deaths must be reported to the Medical Examiner by California state law. Presumed SCDs⁶ were defined using World Health Organization criteria: unexpected out-of-hospital cardiac arrest

death occurring within 1 hour of symptom onset (event witnessed) or within 24 hours of last being observed alive and symptom-free (unwitnessed).²⁸

The initial POST SCD cohort (February 2011 to March 2014) included all incident presumed SCDs countywide, totaling 525 cases. The extended cohort (March 2014 to March 2023) included cases based on medical examiner call schedule (approximately every third day). Exclusion criteria were terminal illness, end-stage renal disease on hemodialysis, identifiable noncardiac sudden death at presentation (eg, evidence of drug abuse at the scene), or recent admission within 30 days for noncardiac illness or surgery. Because this substudy evaluated premortem cardiac diagnoses, presumed SCDs without evidence of at least 1 documented health care visit in the medical records were excluded. As a control group, we included nonsuicide, nonhomicide, accidental trauma deaths referred for medicolegally mandated autopsy during the study period.²⁹

POSTMORTEM EXAMINATIONS AND ADJUDICATION OF CAUSES OF PRESUMED SCDs. All presumed SCDs underwent comprehensive postmortem examinations following Association for European Cardiovascular Pathology guidelines,³⁰ including full autopsy with gross and microscopic evaluation and toxicological analysis by a board-certified forensic pathologist with review by a cardiovascular pathologist. Autopsy protocol was consistent throughout the study period, with the addition of COVID testing performed in all cases during the peak pandemic period of July 2020 to June 2021 and selected cases after this period. A multidisciplinary committee consisting of the medical examiner, cardiac pathologist, neurologist, and 2 cardiologists/cardiac electrophysiologists reviewed premortem medical records, forensic investigator reports, emergency medical service records, prescriptions, autopsy, toxicology, and histology findings to adjudicate a single cause of arrhythmic (potentially rescuable with ICD) or nonarrhythmic death. Detailed autopsy procedures and adjudication criteria for arrhythmic death are described in the [Supplemental Appendix](#).

SENSITIVITY AND POSITIVE PREDICTIVE VALUE OF DIAGNOSED RFs FOR ARRHYTHMIC DEATHS. All available premortem medical records from the 10 San Francisco County adult hospitals and emergency medical service agencies were reviewed to identify diagnosed conventional RFs for SCD. We assessed the following RFs: left ventricular ejection fraction (LVEF) $\leq 35\%$, documented on any cardiac imaging study before death; HF, defined by a documented diagnosis in medical records (including preserved

and reduced ejection fraction) and/or LVEF $\leq 35\%$ by imaging; prior MI, defined by a documented history of MI; and syncope, defined as a documented episode of transient loss of consciousness or diagnosis of syncope (see the [Supplemental Methods](#) for more details).

To assess the real-world performance of these RFs, we calculated sensitivity (ie, prevalence) and positive predictive values of these diagnosed RFs to identify countywide arrhythmic deaths potentially rescuable with defibrillator among total presumed SCDs (calculation methods described in [Supplemental Figure 1](#)). We were unable to assess specificity or likelihood ratios because the prevalence of the RFs in the underlying population (San Francisco County) is unknown. We also assessed GDMT (definition and assessment criteria in [Supplemental Methods](#)) and ICD use among arrhythmic deaths with diagnosed RFs. For patients with LVEF $\leq 35\%$, contemporaneous GDMT was defined as angiotensin-converting enzyme inhibitors/angiotensin receptor blocker, beta-blocker, and aldosterone antagonist use because some of the cohort predated sodium-glucose cotransporter 2 inhibitor use as part of GDMT.

ASSESSMENT OF OCCULT CARDIAC PATHOLOGY.

To evaluate occult cardiac pathology among presumed SCDs without diagnosed RFs, we performed a hierarchical classification of postmortem cardiac findings. In those without RFs, we first ascertained occult MI and DCM as analogs to diagnosed SCD RFs of prior MI or HF, respectively. Occult MI was defined as histopathological evidence of healed MI (nonacute myocardial gross scar or microscopic replacement fibrosis attributed to myocardial ischemia, including cases with and without obstructive CAD [the latter defined as MI with nonobstructive coronary arteries]). CAD was defined as $\geq 50\%$ luminal diameter ($= 75\%$ area) stenosis in ≥ 1 major artery, confirmed independently by 2 pathologists (A.J.C. and E.M.), or prior bypass surgery or percutaneous coronary intervention. Occult DCM was defined as a left ventricular (LV) short-axis diameter ≥ 3.5 cm and heart weight Z-score ≥ 1.0 , where Z-score reflected normalized heart weight based on sex, age, height, and body weight derived from 27,645 reference autopsies (eg, Z-score of “+1” = 1 SD above expected).³¹ These thresholds were based on analysis of our control trauma deaths and corresponded to the 90th and 85th percentiles for LV short-axis diameter and heart weight Z-score, respectively, indicating hearts with evident dilation and increased mass relative to the background population, as well as expert cardiovascular pathologist assessment.³²

TABLE 1 Clinical Characteristics of Study Population

	Presumed SCD			Trauma Death (n = 130)	P Value	
	Total (N = 877)	Arrhythmic (n = 513)	Nonarrhythmic (n = 364)		Presumed SCD vs Trauma	Arrhythmic vs Nonarrhythmic
Demographics						
Age, y	59.9 ± 14.5	61.1 ± 14.3	58.1 ± 14.7	57.2 ± 18.4	0.10	0.002
Male	640 (73)	405 (79)	235 (65)	96 (74)	0.80	<0.001
Race or ethnicity					0.004	0.011
Asian	173 (20)	112 (22)	61 (17)	40 (31)		
Black	142 (16)	65 (13)	77 (21)	12 (9)		
Hispanic/Latino	59 (7)	33 (6)	26 (7)	15 (12)		
White	478 (55)	288 (56)	190 (52)	61 (47)		
Other	25 (3)	15 (3)	10 (3)	2 (2)		
Body mass index, kg/m ²	28.9 ± 8.1	29.8 ± 8.7	28.3 ± 7.9	26.7 ± 5.4	<0.001	0.007
Medical history						
Left ventricular ejection fraction data available	214 (24)	139 (27)	75 (21)	14 (11)	<0.001	0.03
Electrocardiogram data available	445 (51)	256 (50)	189 (52)	38 (29)	<0.001	0.60
Hypertension	461 (53)	287 (56)	174 (48)	29 (22)	<0.001	0.02
Diabetes	199 (23)	121 (24)	78 (21)	13 (10)	<0.001	0.50
Dyslipidemia	251 (29)	176 (34)	75 (21)	23 (18)	0.009	<0.001
Coronary artery disease	143 (16)	105 (21)	38 (10)	11 (9)	0.02	<0.001
Tobacco use	337 (38)	184 (36)	153 (42)	23 (18)	<0.001	0.06
Alcohol abuse	211 (24)	102 (20)	109 (30)	15 (12)	0.001	<0.001
Illicit drug use	139 (16)	52 (10)	87 (24)	14 (11)	0.10	<0.001
Values are mean ± SD or n (%). SCD = sudden cardiac death.						

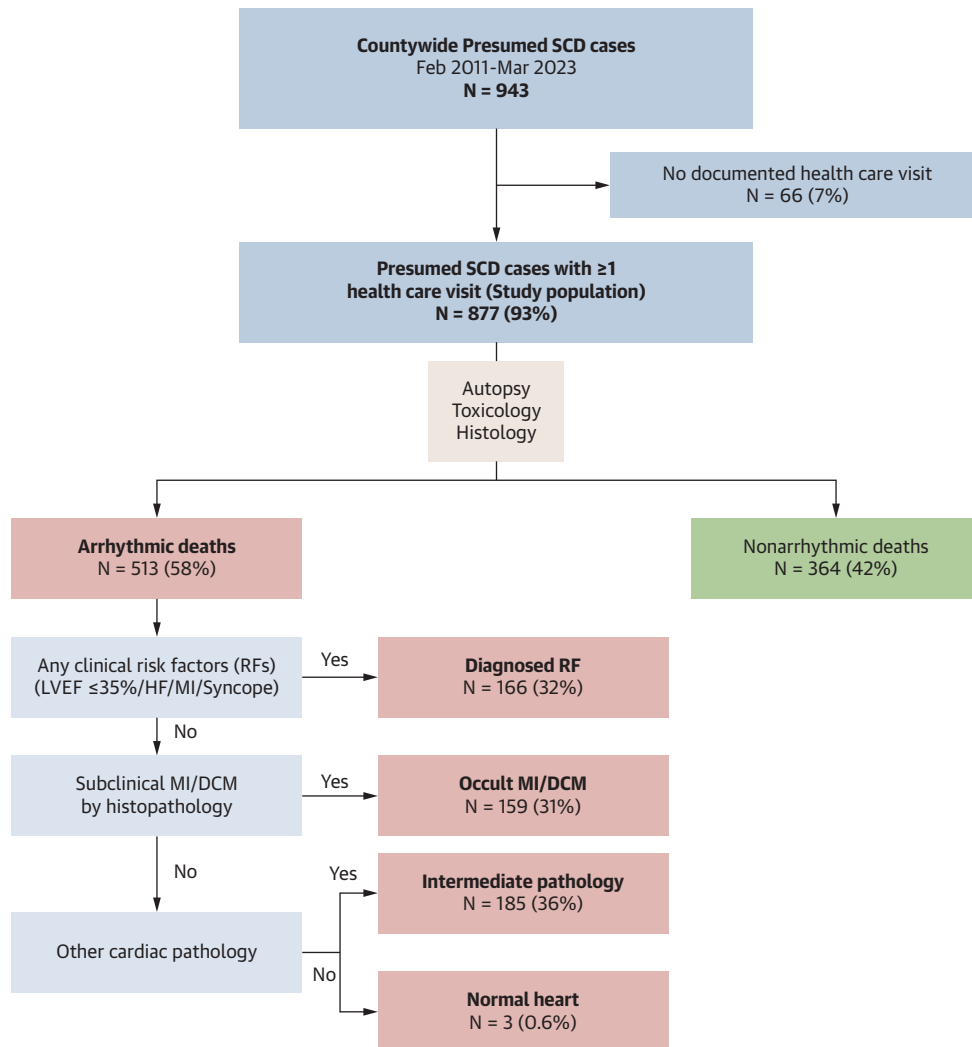
For the remaining cases without diagnosed RFs or occult MI or DCM, we ascertained intermediate cardiac pathology including obstructive CAD (defined as above) without healed MI, LV hypertrophy (LVH; any LV wall thickness >1.5 cm), LV dilation without increased cardiac mass (LV short-axis ≥3.5 cm and heart weight Z-score <1.0), increased cardiac mass without dilation (LV short-axis <3.5 cm and heart weight Z-score ≥1.0), and non-MI LV fibrosis. Cases without any of the above pathologies were classified as normal hearts. Detailed definitions of occult cardiac pathology are described in the [Supplemental Methods](#).

STATISTICAL ANALYSIS. We compared trauma control subjects with 3 mutually exclusive subgroups of arrhythmic deaths: those with diagnosed RFs, those with occult MI or DCM, and those without RFs or occult MI or DCM (intermediate cardiac pathology). We present data as mean ± SD or median (IQR) as appropriate for continuous variables and as proportions with percentages for categorical variables. We used the unequal-variance *t* tests for continuous variables and Pearson's chi-square test or Fisher's exact test for categorical variables as appropriate. We conducted comparisons for multiple groups by the analysis of variance, Kruskal-Wallis test, or Z-test as

appropriate, followed by Bonferroni correction. We compared differences in the burden of autopsy findings using logistic regression models for categorical variables and linear models for continuous variables, with adjustment for age, sex, and race (except heart weight Z-scores as already standardized). For fibrosis analyses, we used log-transformed values. All tests were 2-sided, and we considered *P* < 0.05 to be statistically significant. All analyses were conducted using SPSS version 29 (SPSS Inc).

RESULTS

Between February 2011 and March 2023 in San Francisco County, 943 presumed SCDs underwent the comprehensive postmortem protocol. Of these, 877 subjects (93%) had at least 1 documented health care visit and defined the study population (mean age: 60 ± 15 years; 73% men) ([Table 1](#)); 513 (58%) had autopsy-defined arrhythmic causes and 364 (42%) nonarrhythmic causes ([Figure 1](#)). We also included 130 trauma control deaths referred for autopsy during the study period. Trauma control subjects were similar in age and sex distribution as total presumed SCDs ([Table 1](#)). Leading arrhythmic causes of presumed SCD were acute or healed MI (39%) and non-ischemic cardiomyopathy (19%); leading noncardiac

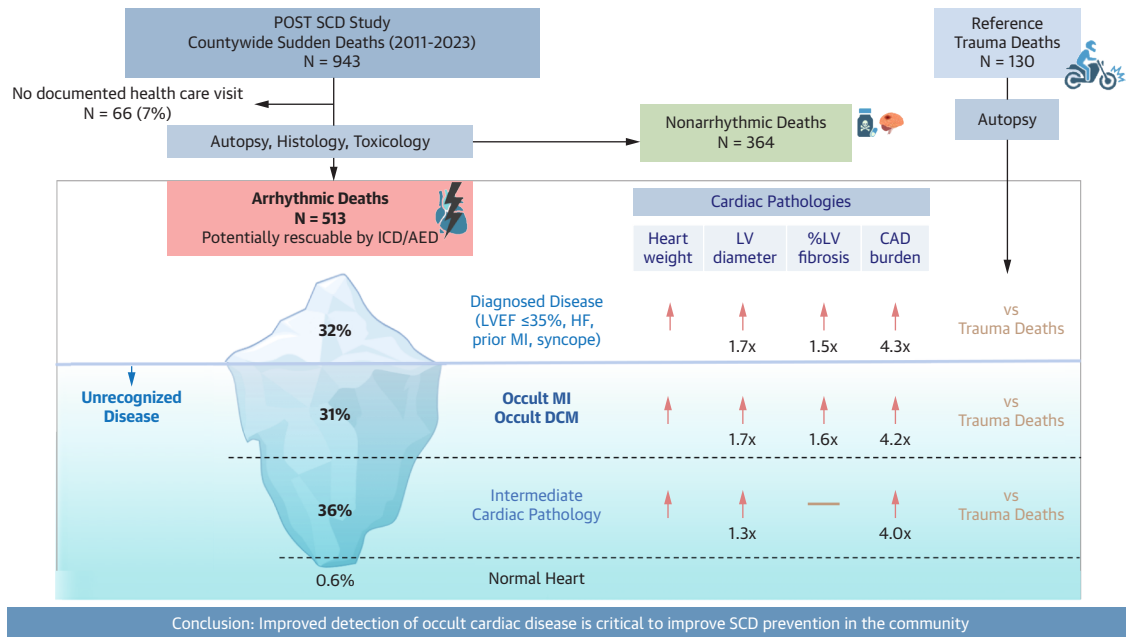
FIGURE 1 Hierarchical Classification of Countywide Presumed SCDs: Identifying Arrhythmic Deaths with Diagnosed Risk Factors and Occult MI or DCM

Study flow diagram showing case selection and hierarchical classification of 877 out-of-hospital cardiac arrest death cases meeting World Health Organization SCD definition (ie, presumed SCDs) with documented health care visits in San Francisco County (February 2011 to March 2023). Among 877 presumed SCDs, 513 (58%) were classified as arrhythmic deaths (potentially rescuable with implantable cardioverter-defibrillator) and 364 (42%) as nonarrhythmic deaths based on systematic postmortem investigations, including autopsy, toxicology, and histology. Arrhythmic deaths are then shown categorized in a hierarchical fashion by presence of diagnosed risk factors (32%), occult MI or DCM (31%), and intermediate pathology (36%). DCM = dilated cardiomyopathy; HF = heart failure; LVEF = left ventricular ejection fraction; MI = myocardial infarction; RF = risk factor; SCD = sudden cardiac death.

causes were occult overdose (31%) and sudden neurological death³³ (15%) (Supplemental Table 1). Subjects defined as arrhythmic deaths were older (mean age: 61 vs 58 years; $P = 0.002$) and more likely to be men (79% vs 65%, $P < 0.001$) than nonarrhythmic deaths. Hypertension and dyslipidemia were more common in arrhythmic deaths, whereas

tobacco, alcohol, and illicit drug use were more common in nonarrhythmic deaths (Table 1).

SENSITIVITY AND POSITIVE PREDICTIVE VALUE OF DIAGNOSED RFs FOR IDENTIFICATION OF ARRHYTHMIC DEATHS. Among 877 presumed SCDs, 247 subjects (28%) had ≥ 1 diagnosed RF, whereas of

CENTRAL ILLUSTRATION Diagnosed and Occult Cardiac Disease Among Community Sudden Deaths at Autopsy

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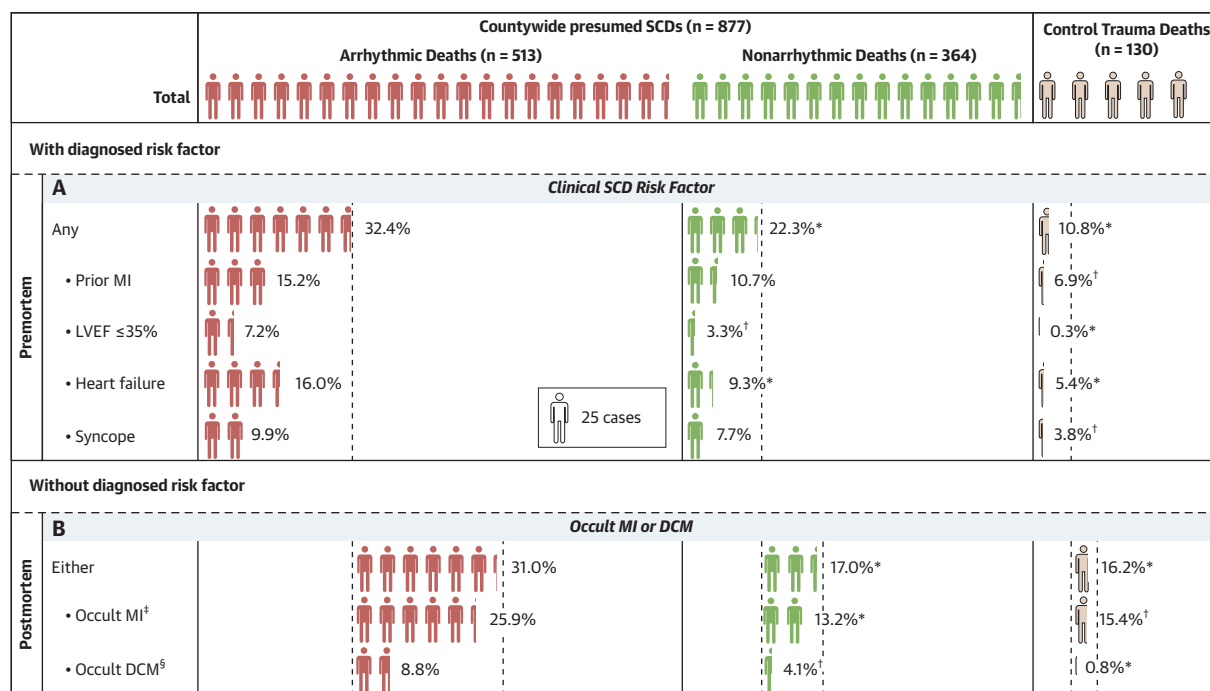
This study performed systematic autopsy with histopathology and toxicology on 943 unselected countywide sudden deaths in San Francisco (2011-2023). After excluding 364 nonarrhythmic deaths (eg, overdose, stroke), 513 autopsy-defined arrhythmic deaths with ≥1 documented health care visits were identified. Among these, only 32% (n = 166) had diagnosed conventional SCD risk factors (LVEF ≤35%, heart failure, prior MI, or syncope). The remaining two-thirds without diagnosed disease were stratified into 31% (n = 159) with occult MI or DCM, 36% (n = 185) with intermediate cardiac pathology, and 0.6% (n = 3) with normal hearts. Comparison with 130 trauma control deaths demonstrated that all arrhythmic death subgroups (aside for the normal heart group) had significantly elevated heart weight, LV diameter, myocardial fibrosis burden (in diagnosed and occult disease group), and CAD burden vs control subjects. These findings reveal that substantial cardiac pathology exists even among those without recognized risk factors or occult MI or DCM, highlighting the critical need for improved detection for community SCD prevention. AED = automated external defibrillator; CAD = coronary artery disease; DCM = dilated cardiomyopathy; ICD = implantable cardioverter-defibrillator; MI = myocardial infarction.

513 arrhythmic deaths, 166 subjects (32%) had ≥1 RF, yielding a 32% countywide sensitivity of diagnosed RFs for predicting arrhythmic death (Central Illustration, Figure 2). This sensitivity was similar by sex and race (Supplemental Table 2). The sensitivity of individual RFs for arrhythmic death ranged from 7.2% for LVEF ≤35% to 16.0% for HF. Although RFs were significantly more prevalent among arrhythmic deaths (32%) than nonarrhythmic deaths (22%, $P = 0.001$) or trauma control subjects (11%, $P < 0.001$), the positive predictive value of any RF for identifying arrhythmic cause among sudden deaths in the community was 67% (Figure 2). The positive predictive value trended higher in men (70% vs 59%, $P = 0.08$) but was similar by race (Supplemental Figure 2). Of the 78 arrhythmic deaths with a

history of MI, 51 (65%) had MI as the cause of sudden death by autopsy (Supplemental Table 1).

Among 37 arrhythmic deaths with documented LVEF ≤35%, only 19% were prescribed all components of contemporaneous GDMT for HFrEF, and 3 (8%) were implanted with ICDs. In individuals with prior MI, statin and antithrombotic use were both 64% (Supplemental Table 3). In total, 5 decedents with ICDs still died suddenly: 3 attributed to device problem identified on postmortem interrogations and 1 to shock-refractory ventricular tachycardia.

ARRHYTHMIC DEATHS WITH OCCULT MI OR DCM. Of the 347 arrhythmic deaths without diagnosed RFs (LVEF ≤35%, HF, MI, syncope), 159 (46%) had occult MI or DCM detected on autopsy. Prevalence of occult MI or DCM among all arrhythmic deaths was

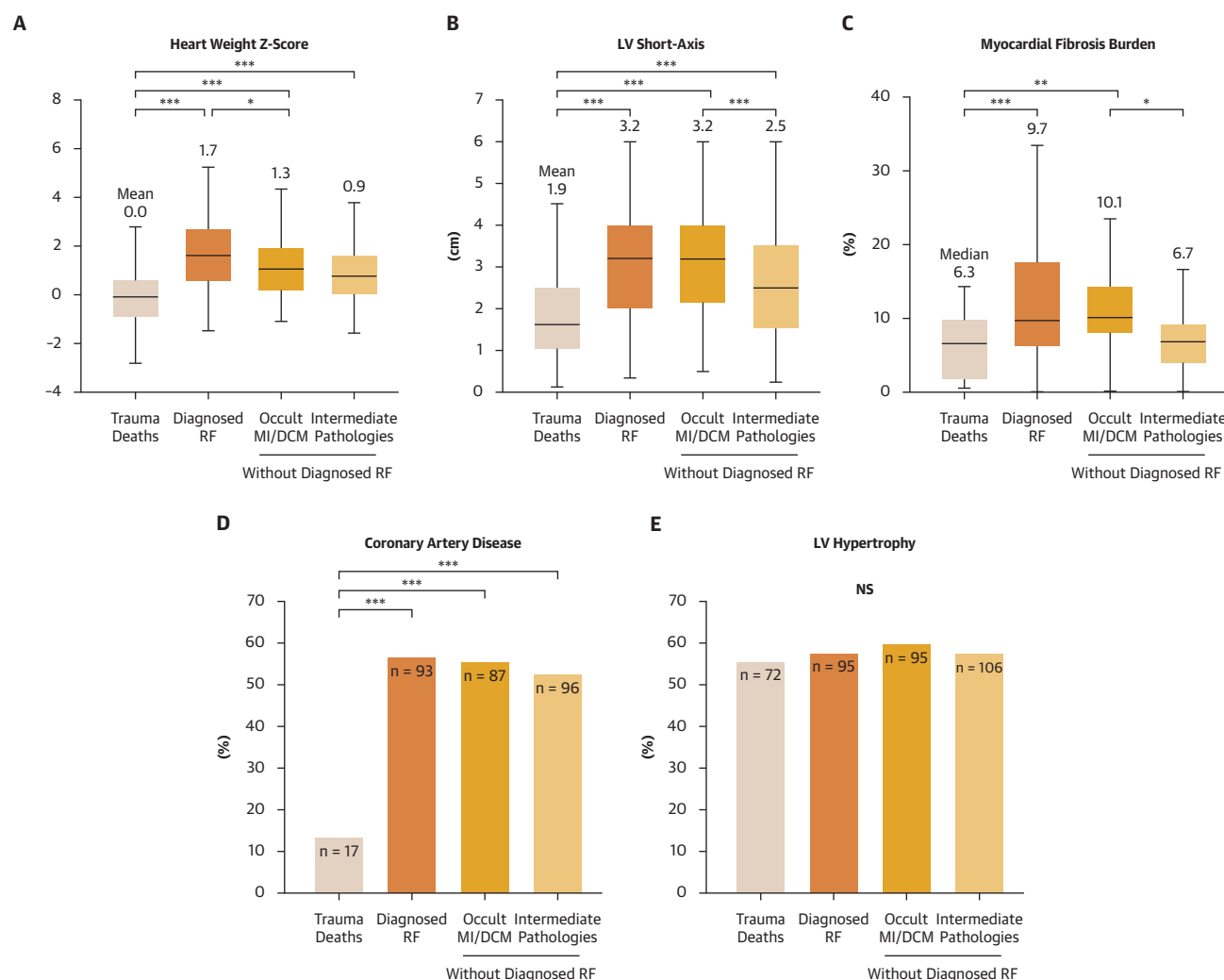
FIGURE 2 Prevalence of Diagnosed SCD Risk Factors and Occult MI or DCM Among Countywide Presumed SCDs and Trauma Deaths

Prevalence of SCD risk factors diagnosed premortem and occult MI or DCM diagnosed postmortem in 513 arrhythmic deaths, 364 nonarrhythmic deaths, and 130 control trauma deaths, the last representing the estimated community prevalence of cardiac pathologies. (A) Clinical SCD risk factors (LVEF $\leq 35\%$, heart failure, prior MI, and syncope) were diagnosed before death in 166 (32.4%) arrhythmic deaths. (B) Histopathologic occult (ie, not diagnosed before death) MI or DCM was identified in 159 (31.0%) arrhythmic deaths by autopsy. * $P < 0.01$ vs arrhythmic deaths. † $P < 0.05$ vs arrhythmic deaths. ‡Histopathologic evidence of healed MI without diagnosed clinical SCD risk factor. §Left ventricular short-axis diameter ≥ 3.5 cm and heart weight Z-score ≥ 1.0 without diagnosed SCD risk factor.

Abbreviations as in [Figure 1](#).

31% (159 of 513), significantly higher than among nonarrhythmic deaths (17%, $P < 0.001$) ([Figure 2](#)). Of those with undiagnosed MI or DCM, 72% had MI alone, 16% had DCM, and 12% had both occult MI and DCM ([Supplemental Figure 3](#)). The distribution of autopsy-defined causes was similar in decedents with diagnosed RFs and those with occult MI or DCM ([Supplemental Table 4](#)). Severity of cardiac pathology in arrhythmic deaths with occult MI or DCM was largely similar to those with diagnosed RFs, including LV short-axis diameter, and burden of CAD, myocardial fibrosis, and LVH. However, heart weight was significantly greater in cases with diagnosed RFs (Z-score: 1.7 vs 1.2; $P = 0.016$) ([Figure 3](#)). The burden of cardiac pathology in arrhythmic deaths with diagnosed RFs and those with occult MI or DCM was significantly higher than that of trauma control subjects, except for LVH ([Figure 3](#)). Arrhythmic deaths with occult DCM also had significantly higher interstitial/perivascular fibrosis burden than trauma control deaths ([Supplemental Table 5](#)).

INTERMEDIATE CARDIAC PATHOLOGY IN ARRHYTHMIC DEATHS WITHOUT RFs, OCCULT MI, OR DCM. Of the remaining 188 arrhythmic deaths without diagnosed RFs or occult MI or DCM (36.6% of arrhythmic deaths), only 3 (0.6%) had normal hearts and the remainder had an intermediate degree of cardiac pathology ([Supplemental Figure 4](#)). Although these deaths did not meet MI or DCM autopsy criteria and had lower levels of fibrosis than those that did ([Supplemental Table 5](#)), the mean heart weight Z-score, mean LV diameter, and burden of significant CAD were all significantly higher than trauma control subjects (Z-score: 0.9 vs 0.0; LV diameter: 2.5 ± 1.2 cm vs 1.9 ± 1.1 cm; CAD: 52% vs 13%; all $P < 0.001$) ([Figure 3](#)). Trauma death heart weights reflected a normal distribution ([Supplemental Figure 5](#)), supporting the validity of trauma deaths as reflective of the reference population. Finally, the distribution of underlying causes of sudden death was similar across arrhythmic death subgroups ([Supplemental Table 4](#)).

FIGURE 3 Comparison of Cardiac Pathology among Arrhythmic Death Subgroups and Trauma Controls

The burden of various cardiac pathologies across arrhythmic death subgroups and trauma deaths as reference representing the background cardiac pathology in the general population. (A) Heart weight Z-score, representing the SD from expected heart weight based on sex, age, height, and body weight. Control trauma deaths had a Z-score of zero, indicating expected heart weight. (B) LV short-axis diameter. (C) LV fibrosis burden. (D) Prevalence of coronary artery disease ($\geq 50\%$ diameter = 75% area stenosis, or any previous coronary intervention). (E) Prevalence of LV hypertrophy (any LV compact wall thickness >1.5 cm). Notably, all pairwise comparisons for LV hypertrophy were nonsignificant. Brackets indicate intergroup comparisons with stars above indicating the significance of intergroup comparisons. (A-C) Boxes represent IQRs, whereas central lines indicate the median values. Error bars represent 5% to 95% ranges. *** $P < 0.001$, ** $P < 0.01$, * $P < 0.05$. All other comparisons were nonsignificant. LV = left ventricular; NS = nonsignificant; Abbreviations as in Figure 1.

CLINICAL CHARACTERISTICS OF ARRHYTHMIC DEATH SUBGROUPS. Table 2 shows the clinical characteristics of the 3 arrhythmic death subgroups (ie, diagnosed RF, occult MI or DCM, and neither). The group without RF or occult MI or DCM was significantly younger (mean age: 57 years) than both the diagnosed RF group (64 years) and occult MI or DCM group (63 years; all $P < 0.001$). LVEF and electrocardiogram data were less often available in those

without diagnosed RFs (LVEF: 13%; electrocardiogram: 36%) compared with the diagnosed RF group (LVEF: 57%; electrocardiogram: 78%). By definition, the subjects defined as arrhythmic deaths with diagnosed RFs had more documented cardiovascular and comorbid medical conditions, but the remaining arrhythmic death subjects still had a substantial prevalence of other documented cardiovascular RFs including hypertension (47%), diabetes (20%), and

TABLE 2 Clinical Characteristics of Arrhythmic Deaths With Diagnosed vs Occult Disease

	Arrhythmic Deaths (n = 513)				P Value ^b
	Diagnosed Risk Factors ^a (n = 166)	Occult Myocardial Infarction or Dilated Cardiomyopathy (n = 159)	Intermediate Pathology (n = 185)	Normal Heart (n = 3)	
Demographics					
Age, y	64.3 ± 14.6	62.6 ± 12.9	56.9 ± 14.4 ^c	65.0 ± 13.1	<0.001
Male	128 (77)	127 (80)	147 (80)	3 (100)	0.60
Race or ethnicity					0.20
Asian	36 (22)	37 (23)	39 (21)	0 (0)	
Black	30 (18)	10 (6)	25 (14)	0 (0)	
Hispanic/Latino	12 (7)	10 (6)	11 (6)	0 (0)	
White	85 (51)	96 (60)	104 (56)	3 (100)	
Other	3 (2)	6 (4)	6 (3)	0 (0)	
Body mass index, kg/m ²	30.4 ± 9.6	30.0 ± 8.5	29.3 ± 8.1	25.5 ± 7.2	0.60
Medical history					
Left ventricular ejection fraction data available	94 (57)	21 (13) ^c	24 (13) ^c	0 (0)	<0.001
Electrocardiogram data available	129 (78)	53 (33) ^c	71 (38) ^c	3 (100)	<0.001
Hypertension	125 (75)	85 (54) ^c	77 (42) ^c	0 (0) ^c	<0.001
Diabetes	53 (32)	38 (24)	30 (16) ^c	0 (0)	0.003
Dyslipidemia	81 (49)	41 (26) ^c	54 (29) ^c	0 (0)	<0.001
Coronary artery disease	76 (46)	19 (12) ^c	10 (5) ^c	0 (0)	<0.001

Values are mean ± SD or n (%). ^aIncludes decedents with documented history of left ventricular ejection fraction ≤35%, heart failure, myocardial infarction, and syncope. ^bP value is for test for difference across all groups. ^cP < 0.05 vs diagnosed disease.

dyslipidemia (28%). The mean body mass index was similar across the 3 groups (29–30 kg/m² in each (Table 2).

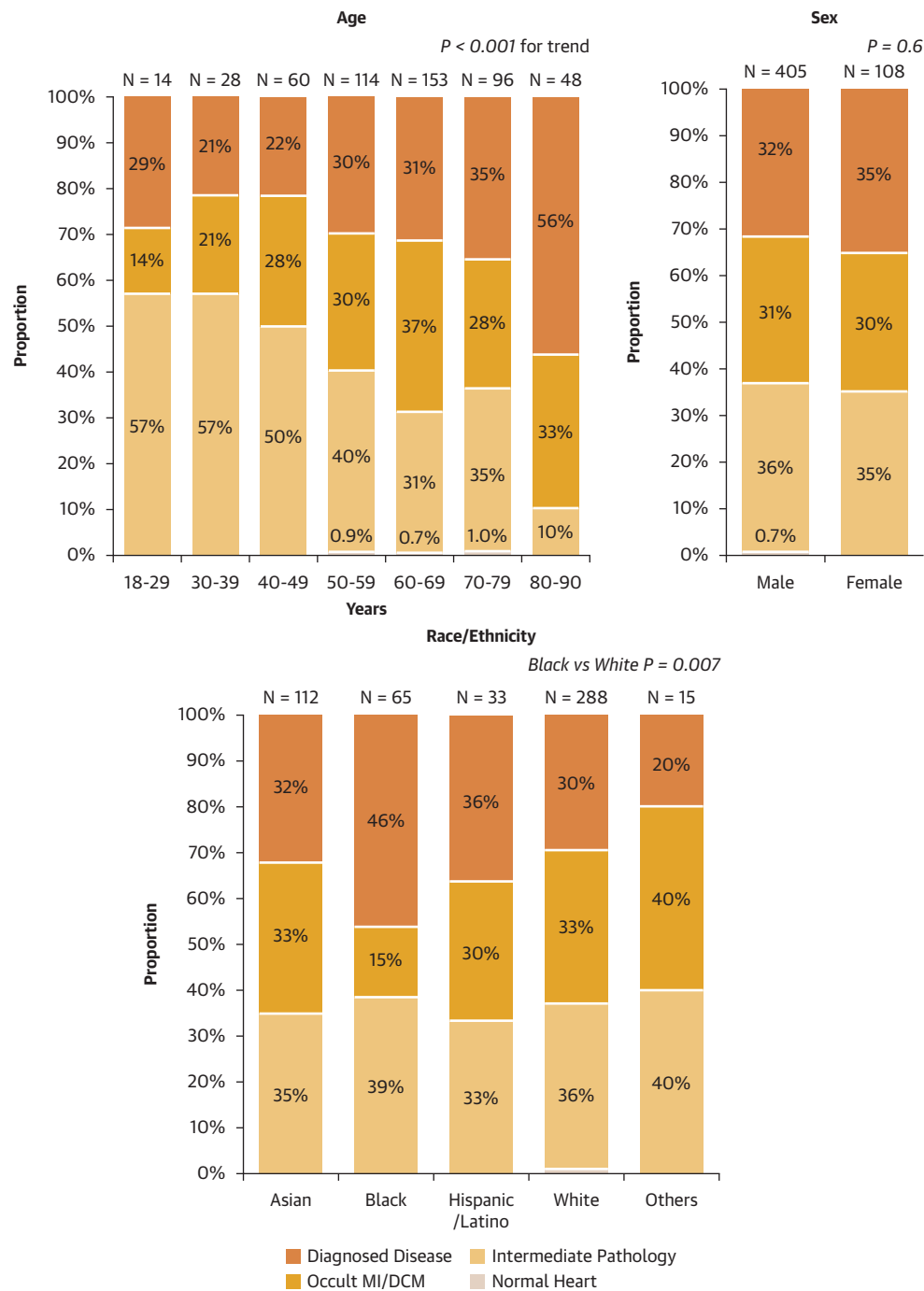
SENSITIVITY AND SUBGROUP ANALYSES. Because syncope is distinct from MI and HF in sudden death prevention strategies and may be nonspecific, we assessed the effect of removing syncope from the evaluated RFs. Of the 51 decedents with history of syncope, 35 (68%) had syncope without concomitant history of MI or HF (Supplemental Figure 6). As such, decedents with only syncope represented 15% of the 247 decedents with recognized RFs, and excluding syncope from the evaluated RFs decreased the proportion of decedents with diagnosed RFs from 32% to 26% and increased the proportion with intermediate cardiac pathology from 36% to 40% (Supplemental Figure 6).

With respect to classification of occult disease, we examined the effect of modulating the heart weight threshold to define DCM. The distribution of occult MI or DCM is shown in Supplemental Figure 4, with most occult cases harboring occult MI (84%). Changing the heart weight Z-score threshold for DCM from the original of 1 to 0.5 (most sensitive) or 2 (least sensitive) changed the yield of detected occult DCM from 5.1% (Z-score of 1) to 7.4% (Z-score of 0.5) or 2.5% (Z-score of 2) and the yield of total detected

occult MI or DCM from 31% (Z-score of 1) to 33% (Z-score of 0.5) or 28% (Z-score of 2) (Supplemental Figure 4).

We also performed stratified analyses by age and race or ethnicity. Diagnosed RFs and occult MI or DCM cases were less prevalent in younger than older decedents ($P < 0.001$ for trend) (Figure 4). However, we found few normal hearts irrespective of decade, including none in those aged <50 years. Race or ethnicity stratified analyses were notable for a higher proportion of diagnosed RFs at 46% among Black decedents and a lower proportion of occult MI or DCM (Figure 4).

TEMPORAL ANALYSES. We evaluated 3 separate time periods: initial study period (February 2011 to March 2014), prepandemic period (March 2014 to 2019), and pandemic period (2020 to March 2023) of the extended cohort. As outlined in Methods, after the initial study period that captured all presumed SCDs, sampling in the extended cohort was based on medical examiner call schedule (approximately every third day). After the initial study period, decedents in the extended cohort were on average 6 years younger ($P < 0.001$) and more often men ($P = 0.017$) (Supplemental Figure 7). The proportion of decedents with diagnosed RFs decreased after the initial study period (37%) compared with the

FIGURE 4 Distribution of Arrhythmic Deaths With Clinical Risk Factors, Occult MI or DCM, and Intermediate Cardiac Pathology by Age, Sex, and Race or Ethnicity

Hierarchical classification of 513 arrhythmic deaths by age, sex, and race or ethnicity. Diagnosed disease prevalence increased significantly with age (14%-56%, $P < 0.001$ for trend) but did not differ by sex ($P = 0.60$). Race or ethnicity analysis demonstrated higher proportion of decedents with diagnosed risk factors among Black decedents (46% vs 30% White, $P = 0.007$). Abbreviations as in [Figure 1](#).

prepandemic (24%) and pandemic periods (30%; $P = 0.022$ for difference across the 3 groups), whereas the proportion of decedents with occult MI or DCM was similar across the 3 periods. There was also an increase in deaths with intermediate cardiac pathology in the prepandemic and pandemic periods (28% vs 51% and 40%, respectively; $P < 0.001$).

DISCUSSION

In this 12-year countywide study of presumed SCDs refined to arrhythmic and nonarrhythmic causes by autopsy, conventionally used RFs for SCD (LVEF $\leq 35\%$, HF, prior MI, and syncope) were identified premortem in only one-third of community sudden deaths attributable to arrhythmia (ie, potentially rescuable with defibrillator). Thus, two-thirds of community arrhythmic deaths were effectively “missed” by real-world SCD risk stratification, yet one-half had undiagnosed MI or DCM and the other one-half still had substantial cardiac pathology including significantly more CAD and LV dilation than trauma control deaths. In sum, our population-based postmortem study in a diverse urban U.S. setting indicates that the contemporary real-world SCD victim has significant undiagnosed cardiac disease and thus would not be captured by current sudden death guidelines that depend on recognized HF or MI. To move the needle against the global sudden death epidemic in the modern era, prioritizing identification of “silent disease” in this vulnerable population is critical.

The first step in SCD prevention is to recognize conditions that convey risk of lethal arrhythmia. Prior studies have reported many individuals die suddenly without diagnosed cardiac disease or RFs for sudden death (Table 3)^{13,14,24,34,35}; however, these studies largely presumed arrhythmic causes without autopsy^{13,14,36} and thus were confounded by the inclusion of noncardiac sudden deaths that have fewer cardiac RFs. As a result, the actual yield of diagnosed RFs for arrhythmic death in the community has not been precisely determined. Our study refined presumed SCDs in the community to those with arrhythmic causes on autopsy and demonstrated that only one-third (32%) of over 500 arrhythmic deaths had a known diagnosed RF for SCD premortem. Our goal was to include major RFs that could have led to targeted diagnostics or therapeutics to prevent sudden death. Thus, we included RFs that may be viewed as nonspecific, such syncope and HF with preserved ejection fraction, because these may have distinct significance in a cohort of sudden deaths.

Notably, the proportion of decedents with diagnosed disease would be even lower had our definition of RFs only included reduced ejection fraction or MI, conditions that have associated primary prevention guidelines for sudden death.

The unexpectedly low prevalence of diagnosed RFs among community sudden deaths suggests sudden deaths are prevented in patients with diagnosed RFs and is consistent with published data demonstrating improved sudden death risk in populations with identified HF.⁵ Although this may be a reason for optimism, subgroup analyses suggested that not all populations may benefit uniformly from conventional prevention strategies. First, our data demonstrate that these commonly used RFs likely have a lower utility in distinguishing arrhythmic risk in women. This is consistent with our previous work demonstrating that women have a distinct profile of arrhythmic death compared with men, with less due to CAD; thus, women may need distinct strategies to mitigate risk.³⁷ Second, Black decedents had a significantly higher proportion of diagnosed RFs (46%) than the remainder of decedents (30%), which raises the possibility of disparities in the optimal implementation of therapies that prevent SCD in those with recognized disease. These findings may collectively help explain why Black individuals suffered the highest incidence of arrhythmic death in POST SCD,²⁰ especially Black women, at 2.5 times higher than their White counterparts.³⁷

However, our data suggest that the most significant gap is how to prevent sudden death in decedents with unrecognized disease, representing two-thirds of total arrhythmic SCDs. Although occult disease has been reported in autopsied SCDs, studies have focused on decedents without diagnosed CAD or cardiomyopathy in a distinct, Northern European population^{24,34} or only ischemic causes^{24,35,38} (Table 3), which limit generalizability to public health efforts directed at *all* persons in the community. As we demonstrate in the companion paper published in the Journal, we found that MI accounts for only 41% of autopsy-confirmed SCD.³⁹ In our study, we used comprehensive postmortem investigation in adult victims up to age 90 to assess all detectable cardiovascular pathology (ie, macro- and microscopic disease) and their ultimate contribution to community sudden deaths. In arrhythmic death victims without known diagnosed RFs, we found that one-half had occult MI or DCM. Notably, these cases had a similar severity of cardiac pathology (ie, burden of CAD, LVH, fibrosis, and LV dilation) and causes as cases with diagnosed disease and thus represent a high

TABLE 3 Community SCA/SCD Studies: Diagnosed and Occult Disease

First Author/ Study, Year	Location/Period	Study Type	Study Population	Study Size and Autopsy Rate	Control Group
de Vreede-Swagemakers, 1997 ¹²	The Netherlands, Maastricht area, 1991-1994	Prospective nonautopsy cohort	Area-wide <i>out-of-hospital cardiac arrest including survivors</i> (asymptomatic or ≤ 24 h of symptom onset)	n = 525 127 autopsied (24%)	None
Albert, 2003 ¹³	USA 1976-1998	Prospective nonautopsy cohort	From Nurses' Health Study: <i>presumed SCD in Women</i> (≤ 1 h after symptom onset)	244 of 121,701 nurses Nonautopsy study	Nurses without presumed SCD
Chugh, 2004 ¹⁴	USA, Oregon, Multnomah County 2002-2003	Prospective nonautopsy cohort	Countywide SCA <i>including survivors</i> (presumed cardiac cause witnessed: ≤ 1 h of symptom onset; unwitnessed: ≤ 24 h from last seen normal)	n = 353 41 autopsied (12%)	None
Chugh, 2009 ³⁶	USA, Oregon, Multnomah County 2002-2007	Prospective nonautopsy cohort	Countywide SCA <i>including survivors</i> (presumed cardiac cause witnessed: ≤ 1 h of symptom onset; unwitnessed: ≤ 24 h from last seen normal)	n = 1,568 NR	None
Adabag, 2010 ⁵⁴	USA, Minnesota, Hennepin County 2001-2004	Retrospective autopsy-based cohort	Countywide <i>out-of-hospital cardiac arrest death</i> (≤ 24 h from symptom onset or last seen alive) <i>without known "severe illness"</i>	n = 71 62% of 114 eligible cases, 580 sudden deaths with severe illness excluded, including cardiomyopathy and coronary heart disease	None
Wu, 2016 ⁵⁵	China, Guangzhou, cases referred to Sun Yat-Sen Univ. 1998-2013	Retrospective autopsy-based cohort	<i>Autopsy-defined SCD</i> (witnessed: ≤ 1 h of symptom onset; unwitnessed: ≤ 24 h from last seen normal) <i>referred for autopsy</i>	n = 1,656 Autopsy rate relative to total SCDs NR	None
Junttila, 2016 ³⁴	Finland, Province of Oulu 1998-2012	Prospective autopsy-based cohort	<i>Autopsy-defined SCD</i> (witnessed: ≤ 6 h of symptom onset; unwitnessed: ≤ 24 h from last seen normal) <i>without known heart disease</i>	n = 2,697 NR, sudden deaths not autopsied if attributable to known disease	None
Vähätalo, 2019 ²⁴	Finland, Province of Oulu 1998-2017	Prospective autopsy-based cohort	<i>Ischemic autopsy-defined SCDs</i> (witnessed: ≤ 6 h of symptom onset; unwitnessed: ≤ 24 h from last seen normal) <i>without prior CAD diagnosis</i>	n = 3,122 NR, sudden deaths not autopsied if attributable to known disease	None
Vähätalo, 2021 ³⁵	Finland, Province of Oulu 1998-2017	Prospective autopsy-based	<i>Ischemic autopsy-defined SCDs</i> (witnessed: ≤ 6 h of symptom onset; unwitnessed: ≤ 24 h from last seen normal) aged < 50 y	n = 266 NR, sudden deaths not autopsied if attributable to known disease	None
Hookana, 2023 ³⁸	Finland, Province of Oulu 1998-2017	Prospective autopsy-based	<i>Ischemic autopsy-defined SCDs in women</i> (witnessed: ≤ 6 h of symptom onset; unwitnessed: ≤ 24 h from last seen normal)	n = 888 NR, sudden deaths not autopsied if attributable to known disease	None
Current study	USA, California, San Francisco County 2011-2023	Prospective autopsy-based	From POST SCD study: <i>autopsy-defined arrhythmic sudden death</i> (witnessed: ≤ 1 h of symptom onset; unwitnessed: ≤ 24 h from last seen normal)	n = 513 2011-2014: 97% of presumed SCDs, 2014-2023: incident cases based on medical examiner call schedule ~every third day	Trauma deaths

CAD = coronary artery disease; EMS = emergency medical service; LVEF = left ventricular ejection fraction; LVH = left ventricular hypertrophy; MI = myocardial infarction; NR = not reported; SCA = sudden cardiac arrest; SCD = sudden cardiac death.

Continued on the next page

priority opportunity for SCD prevention because they were unrecognized and naive to targeted prevention efforts.

The remaining one-third of arrhythmic deaths had neither diagnosed disease premortem nor met autopsy criteria for MI or DCM. However, these cases

still had a substantial burden of cardiac pathology, including 4-fold higher levels of CAD and 1 SD heavier and 1 cm more dilated hearts than the background population at risk for SCD (trauma deaths), reflecting an intermediate degree of cardiac pathology. This burden of pathology is particularly striking,

TABLE 3 Continued

Age Range (y) (Mean/Median)	Male (%)	Race or Ethnicity	Data Source	Toxicology Screen	Proportion with Diagnosed Disease (% of Included Decedents)	Occult Disease at Autopsy
20–75 (62)	72	NR	National data, EMS, and medical records	NR	Cardiac diagnosis = 54% Prior MI = 35%	NR
30–55 at enrollment (NR)	0	NR	Self-report, family interview, medical records, death certificate	NR	Cardiac diagnosis = 31%	NR
All ages (69)	57	NR	EMS, medical examiner, medical records, death certificate	Yes, in all autopsied	Conditions associated with SCD in decedents with clinical records/autopsy results (188, of which 41 were autopsied). Not reported whether premortem diagnosis or detected on autopsy: CAD 45% Congestive heart failure/cardiomyopathy 25% No significant associated conditions 37%	
≥35 (67)	65	NR	EMS, medical examiner, and medical records	NR	Of 1,258 with clinical records available: CAD = 46% LVEF ≤ 35% = 12% LVH = 12% Prior MI = 28%	NR
25–60 (50)	86	Asian 1% Black 1% Hispanic 1% White 96%	Family interview, medical records, autopsy data	Yes	CAD = 3%	High-grade CAD = 80% Silent MI = 34% LVH = 58% Interstitial fibrosis = 55%
All ages (38)	81	NR	Autopsy data	NR	NR	NR Only reported autopsy defined causes of death: 42% CAD; 15% sudden unexplained death; 12% myocarditis; 4.7% cardiomyopathy
All ages (62)	79	White 100%	Family interview, medical records, death certificate, autopsy data	Yes, if needed	0% (excluded by design)	Ischemic cause (CAD or MI scar) = 66%–74% Idiopathic myocardial fibrosis = 6.7%
All ages (66)	79	White 100%	Family interview, medical records, death certificate, autopsy data	Yes, if needed	Heart failure = 5.9%	Silent MI = 42%
<50 (44)	90	White 100%	Family interview, medical records, death certificate, autopsy data	Yes, 80%	CAD = 9.8% Prior MI = 6.2%	Of those without CAD diagnosis: MI scar = 34% LVH = 58%
All ages (73)	0	White 100%	Medical records, death certificate, autopsy data	NR	CAD = 26% Prior MI = 6.8%	Of those without CAD diagnosis: silent MI = 34%
18–90 (61)	73	Asian 22% Black 13% Hispanic 6% White 55%	EMS and medical records, forensic and autopsy reports	All cases aged < 75 y and ≥ 75 y if no evident cause	Prior MI = 15% LVEF ≤ 35% = 7% Heart failure = 16% Syncope = 10% ≥1 of the above risk factors: 31%	Occult MI or dilated cardiomyopathy = 31% Intermediate pathology (CAD, LVH, fibrosis, LV dilation) without diagnosed risk factors (see left) = 36% Normal heart = 0.6%

considering this was the youngest group of arrhythmic deaths (mean age: 57 years).

In contrast to autopsy studies of young adult sudden deaths with higher rates of normal hearts or sudden unexplained death,^{26,40,41} we did not identify any normal hearts in decedents aged < 50 years in our countywide cohort. This may be due to differences in selected populations in other studies (eg, excluding deaths with CAD)^{26,41} or our comprehensive methodology to exclude occult overdose in younger victims with apparently normal hearts and to detect gross and histological evidence of disease in such cases. Therefore, prior studies may have

overestimated rates of normal hearts and yield of genetic testing for primary electrical diseases in this population.^{42,43} Our data suggest that most young adult SCD victims have detectable cardiac disease, and expanding the focus past rare familial arrhythmogenic disorders is necessary to prevent the especially tragic outcome of younger sudden deaths in the community.

Although the exact reason(s) for the unrecognized disease in this population is unclear and outside the scope of the study, we hypothesize that these decedents may have had milder symptoms or lower access to health care. Regarding the former,

arrhythmic deaths with occult MI or DCM had lower heart weights compared with those with diagnosed RFs and thus may reflect a population less susceptible to HF or anginal symptoms that prompt evaluation, while still harboring arrhythmic risk. Although it is possible that decedents had unrecognized disease because of a lack of routine health care, we excluded decedents without at least 1 identifiable medical contact before death, thus preventing inflation of occult disease prevalence simply because of inclusion of decedents without access to health care. Moreover, in those without diagnosed sudden death RFs, approximately one-half still had known hypertension, suggesting these decedents may have had a potential route for intervention. Regardless, undiagnosed cardiac disease underpinning SCD risk due to lack of access to care represents a real-world limitation of SCD prevention efforts. This last factor is a critical limitation of studying SCD within prospective cohorts or randomized control trials that by definition enroll individuals connected to care and thus do not reflect health care use in the community. Understanding the health care use patterns of those dying suddenly to identify gaps and ensure access to standard health care may improve detection and management of occult disease and other cardiovascular RFs such as diabetes or hyperlipidemia to ultimately reduce sudden death.

Detection of subclinical cardiovascular disease and identifying high-risk populations for screening should remain a priority. Although population-level screening of subclinical cardiovascular disease has not been shown to be effective,⁴⁴ those studies may have missed high-risk patients without access to care. Furthermore, advancing technology and artificial intelligence, such as electrocardiogram for prior MI and structural heart disease^{45,46} or computed tomography for CAD,^{47,48} may enable new opportunities for increased and earlier detection of cardiovascular disease within the scope of routine care.

Our study also revealed critical gaps in implementation, such as GDMT use among those with diagnosed RFs. Only 19% with HFrEF and approximately 60% with prior MI received complete GDMT.^{49,50} Improved implementation of GDMT represents a modifiable target to reduce arrhythmic death risk and an immediate opportunity for preventing sudden death. Moreover, of 37 arrhythmic death victims with LVEF $\leq 35\%$, only 3 (8%) had received an ICD. We previously reported that one-third of these ICD-eligible cases had no identifiable barriers to implantation,⁹ suggesting systematic underuse rather than patient-specific

contraindications. Even among ICD recipients with HFrEF, however, 2 of 3 died due to ventricular arrhythmia undersensing, highlighting the need for improved device programming and follow-up as well.⁵¹

STUDY LIMITATIONS. First, although it captures the real-world population in our community, it was not designed as a population-based analysis of all individuals at risk for SCD; thus, the utility of the various cardiac pathologies reported (eg, MI or DCM) for population-level risk stratification is unknown. However, we compared the pathology among arrhythmic deaths with that in trauma deaths as an estimate for background community prevalence of heart disease. The validity of this approach is supported by the alignment of heart weights of trauma deaths in our study (mean heart weight Z-score of 0) with heart weights from a large reference autopsy cohort and a normal distribution of heart weights.

Second, during the latter years of the study period, not every incident presumed SCD in San Francisco County was captured, potentially introducing selection bias. Our selection method of cases was intended to be random (cases occurring approximately every third day based on call schedule) and thus still accurately represents causes and burden of underlying disease. However, the autopsied population after the initial study period was younger by ~ 6 years, which may reflect shifts in medical examiner resources because of an increased burden of forensic cases over the study period rather than a decline in age of the average adult victim of SCD. Still, the average age of our SCD cases is among the oldest mean age of modern autopsy studies of SCD, thereby shedding light into the burden of undiagnosed disease in most SCD victims who are middle age and older. Moreover, the number of decedents dying without diagnosed RFs was largely stable during the studied periods (63% in initial study, 76% in pre-pandemic, and 70% in pandemic periods). Our study population also remains the most generalizable to an increasingly diverse U.S. population (Table 3).

Finally, although we identified pathologic substrates that may be detected by current diagnostic methods, the correlation between premortem diagnostic findings and postmortem pathologic findings was not feasible. When possible, sensitivity analyses were performed to demonstrate the potential effect of modulating different criteria (eg, heart weight for the diagnosis of DCM) on outcomes. Although it was not our goal to explicitly identify thresholds that discern risk, the selected phenotypes of occult MI and DCM represent a distinct target population for prevention because they had similar burdens of CAD,

LVH, fibrosis, and LV dilation as those with diagnosed disease. Autopsy has enabled key insights into what is causing sudden deaths, but further work is needed to translate these findings into appropriate methods and modalities to detect occult disease and then to translate evidence-based interventions to prevent sudden death in the community.

CONCLUSIONS

In this 12-year communitywide sudden death study with comprehensive pre- and postmortem investigations, we found that conventional SCD RFs documented premortem identified only one-third of total arrhythmic deaths in the community. Thus, these widely used RFs miss two-thirds of individuals dying suddenly from lethal arrhythmia. Half of these silent arrhythmic deaths had occult MI or DCM by autopsy, whereas the remainder still had substantial cardiac pathologies, including increased heart weight, larger LV diameter, and significant CAD. These findings suggest that improved detection of occult cardiovascular disease is critical in the prevention of sudden death.

While advances in the prevention of sudden death enabled by enhanced premortem rhythm monitoring, imaging modalities, and artificial intelligence provide reason for optimism, these depend on having a well-defined outcome. Our study demonstrates how high-quality SCD research is still contingent on understanding its underlying causes, missed disease, and generating precise phenotypes of sudden death. Thus, we believe autopsy will remain irreplaceable in providing insight into sudden deaths and as a rigorous means for quality assurance and outcomes research; for example, even perimortem documentation of ostensibly lethal rhythms does not guarantee arrhythmic cause.^{33,52} Coupling pre- and perimortem clinical data and autopsy on a population level has the potential to refine our understanding of SCD, from its anatomical origins to the failures of detection that still cost so many lives, and ultimately guide the public health interventions necessary to curtail the “intractable” sudden death epidemic.⁵³

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DATA AVAILABILITY STATEMENT The data underlying this article may be made available permitting suitable request approved by the corresponding author.

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PERSPECTIVES

COMPETENCY IN MEDICAL KNOWLEDGE: This prospective countywide study combined premortem clinical data with postmortem histopathologic findings, revealing that two-thirds of arrhythmic SCDs occurred in individuals without diagnosed SCD RFs but with substantial cardiac disease at autopsy.

COMPETENCY IN PATIENT CARE: Enhanced health care access and targeted cardiac screening for individuals with cardiovascular RFs are needed for improved detection of occult cardiac disease and prevention of sudden death.

TRANSLATIONAL OUTLOOK: Novel screening approaches beyond conventional SCD RFs (LVEF $\leq 35\%$, HF, prior MI, syncope) may better identify the silent population with occult cardiac pathologies at risk for SCD.

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APPENDIX For a shared methodological appendix as well as an expanded Methods section and supplemental tables and figures, please see the online version of this paper.